

ZONE A POWER CHAIN

4S3P pack, common-port balancing BMS, USB-C charge

S1 and S2 both hang off the fused node, in PARALLEL. S1 does NOT switch the lidar -- opening S1 kills the Pi and the motor and leaves the VLP-16 spinning on S2.

*** NEW ON 4S: S2 HANDS THE VLP-16 UP TO 16.8 V ***
On the old 3S assumption that leg never passed 12.6 V. CHECK THE SENSOR'S OWN INPUT RANGE before S2 is ever closed. If it will not take 16.8 V that leg needs a regulator -- the one place going to 4S makes things worse.

